

Early Phase SoAs

Early-phase Schedules of Activities, and the representation issues they raise for USDM.

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Summary

Early-phase / experimental-medicine studies — short, small-population, intensively-sampled, dosing-anchored trials — produce Schedules of Activities (SoAs) whose *cells* carry timing structure that a plain activity×timepoint grid cannot hold. This report catalogues those concrete representation issues and asks, for each, whether the CDISC Unified Study Definitions Model (USDM) can carry it.

It is a companion to the general SoA and Footnote Patterns report, which catalogued twelve SoA patterns and twenty-one footnote categories across 169 protocols and showed USDM carries them all. That scan is **thin on exactly this population**, so this report works the delta: the cell-level demands these early-phase tables make, evidenced by a set of real protocols and, most directly, by the SoA screenshots in `sources/EMP Study.docx`.

The report has three parts: Part 1 fixes the study type and terminology (see the companion note `docs/phase0_definition_and_terms.md`); Part 2 is the protocol set; Part 3 is the issue-by-issue characterisation and USDM assessment.

Status: draft. The five issues are characterised and grounded (Part 3). The USDM verdicts are marked either **traced** (settled against DDF-RA in prior work, cited) or **open** (needs a DDF-RA trace — the current next step). No USDM capability is asserted without a trace.

Part 1 · What these studies are

A **Phase 0 / exploratory study** gives a sub-therapeutic exposure under a reduced preclinical package to gather early PK, PD, mechanistic or imaging data — not safety/efficacy, not therapeutic benefit. In practice the set used here is broader: **early-phase experimental medicine**, which includes conventional Phase 1 clin-pharm PK/DDI studies. Those turn out to be where the SoA representation issues are sharpest.

ClinicalTrials.gov has no "Phase 0" value — these register as **Early Phase 1**, and much pharma early-phase work is simply labelled Phase 1. Full definition, thresholds and terminology: `docs/phase0_definition_and_terms.md`.

Part 2 • The protocol set

The corpus for the analysis is **26 protocols** sourced from ClinicalTrials.gov (real posted protocol documents only). The full chosen/rejected list, with sponsor, indication, SoA form and timing anchor, is `docs/project_protocols.md`. In shape it splits into:

- **Clin-pharm PK / DDI (Eli Lilly / Loxo)** — the four EMP protocols below; conventional crossover PK and drug-drug-interaction designs whose SoAs carry the cell-level structure this report is about.
- **PET receptor-occupancy / radioligand-tracer** studies (e.g. Asceneuron, Pfizer, Takeda, Genentech, GSK) — dose/scan-anchored, some microdose tracer.
- **Pharmacological- and infection-challenge** studies (LPS, endotoxin, alcohol, amphetamine, influenza) — assessments timed relative to a challenge administration.
- **Imaging microdose** studies (fluorescence / optical probes).

The four EMP protocols are the primary evidence set (from `sources/protocols.docx`), all Eli Lilly / Loxo Phase 1 healthy-volunteer PK/DDI:

NCT	Study	What it is
NCT05469126	J2A-MC-GZGM	clarithromycin + LY3502970 DDI — the source of the SoA screenshots ("Example 2")
NCT05176314	—	pirtobrutinib + rosuvastatin DDI
NCT06085482	—	LY3502970 Phase 1
NCT05444556	—	imlunestrant, female healthy participants (crossover)

Part 3 • SoA representation issues and USDM impact

Each issue below is grounded in the GZGM (NCT05469126) SoA and the other EMP screenshots. "USDM status" is **traced** (settled, cited) or **open** (needs a DDF-RA trace).

Issue 1 — Interval / duration activities that cross day boundaries

What the SoA shows. A 24-hour urine collection recorded in 8-hour bins (4-hour in some studies), the bins running across the study-day columns and merged to show it:

-24 to -16h, -16 to -8h, -8 to 0h, 0 to 8h, 8 to 16h, 16 to 24h ... 48 to 56h, 56 to 64h, 64 to 72h

What it demands. Each bin is an activity with an explicit **start and end** — a duration, not a point. A single logical activity spans several day columns, and the interval can **begin before dosing** (-24 to -16h).

USDM status: OPEN. USDM timing is expressed primarily as point references (a Timing relative to a reference event, with a window tolerance). Whether a collection *interval* (start→end as the activity's own extent, distinct from a $\pm$ scheduling window) has a first-class representation — or must be modelled as two bounding timings, or an activity duration — needs tracing against the DDF-RA API and CORE rules. *This is the most likely genuine delta; do not assert until traced.*

Issue 2 — Dose-relative timing in extended-hour notation

What the SoA shows. PK sampling timed in running hours relative to the dose, past 24:

P, 0.5, 1, 2, 4, 6, 8, 12, 16 then 24 (D2), 36 (D2), 48 (D3) ... 120 (D26), 168 (D28), 240 (D31) — "Times are relative to LY dosing"

Dave's note asks the tool to accept 24:00 hr, 36:00 hr, 48:00 hr ... — hours that keep counting rather than resetting to a day+clock.

What it demands. The primary coordinate is the **elapsed time from dosing**; the "(D2)... (D31)" is the calendar day the offset happens to land on — a rendering, not the stored value.

USDM status: traced — handled. A discrete action timed off dosing is a scheduled activity instance with a relative timing offset (see `lessons_learned.md`). The offset is an ISO 8601 duration (PT24H , PT240H), so extended hours are just a larger duration; day-in-parentheses is a display convention over the same stored offset. The open part is presentational (can the authoring tool *enter/show* 240:00 hr), not a model gap.

Issue 3 — "P": a predose anchor in individual cells

What the SoA shows. A literal P inside cells, per procedure, per period — P for the Day-1 Medical Assessment and ECG, P, 0.5, 1, 2 ... heading a PK series, P (24 hr) for a coproporphyrin draw — re-anchoring in Period 2 of the crossover.

What it demands. A reusable relative anchor meaning "immediately before the dose in *this* period", resolving to the correct dose administration in each period.

USDM status: traced — handled (one point to confirm). "Predose" is a Timing of type *Before* relative to the dose administration; relative-timing anchors are the settled case.

The one thing worth confirming in the trace: in a multi-period crossover, that each period's **P** references that period's dose-administration instance rather than a single global anchor.

Issue 4 — Footnote-encoded scheduling semantics

What the SoA shows. Dave's note: *"Heavy usage of footnotes."* The cell shows only **x^b** or **x^c**; the rule is in the footnote:

- **Windows** — *"within ± 1.5 hours of the scheduled time."*
- **Triplicate + borrowed timing** — *"measured in triplicate at predose and 12h, 24h, 48h, 72h and 96h to match PK collection timing"* (cell shows **x^c**).
- **Conditional repeats** — injection-site assessment *"repeated daily until resolution"; "Failure to administer ... will not result in a protocol deviation."*
- **Ordering** — *"If multiple procedures take place at the same time point, the order should be: ECG, vital signs, blood samples."*

What it demands. Timing, repetition count, conditionality and ordering — carried as prose, not as a cell mark.

USDM status: mixed.

- *Windows* — **traced/handled**: timing-window tolerance is an existing primitive.
- *Triplicate / repeat counts* — **open**: whether a repeat count attaches to one activity or expands to N instances needs tracing.
- *Conditional repeats ("until resolution")* — **open**: needs decision/condition + exit-condition logic; trace required.
- *Co-timed ordering* — **open**: ordering *among activities sharing a timepoint* is subtler than the per-instance timeline ordering already known; trace required.

Issue 5 — Parallel timelines that start before the dose

What the SoA shows. A sampling series running as its own timeline, beginning the day before dosing and continuing across it — coproporphyrin **0, 0.5, 1, 2, 4, 6, 8, 12, 16 hr** on Day -1, then **P (24 hr)** on Day 1 — plus the pre-dose urine bins from Issue 1.

What it demands. A **second timeline alongside** the main dose-anchored one, whose zero is not the dose and which spans the dosing moment.

USDM status: traced — likely handled (confirm). Parallel / continuous monitoring is the settled cyclic-sub-timeline-with-exit case (`lessons_learned.md`); a sub-timeline can anchor on any reference event, so a pre-dose-anchored series that spans the dose should follow the same shape. The new angle to confirm in the trace is only that the anchor precedes the main timeline's anchor — probably fine, not asserted.

Where this leaves USDM

Of the five, **two map onto already-traced relative-timing primitives** (Issues 2 and 3) and **one onto the known sub-timeline pattern** (Issue 5). The genuine open questions — the candidate deltas — are **Issue 1 (interval/duration activities with explicit start→end spanning columns)** and **parts of Issue 4 (repeat counts, conditional "until resolution" repeats, co-timed ordering)**. These are the items to trace against DDF-RA (API + CT + CORE rules) before any verdict. This reopens, on firmer ground, the earlier "no USDM delta" conclusion, which was reached on a narrower framing (challenge anchor + narrative round-trip only).

References

- Companion report: [protocol_soa_patterns/docs/reports/soa_patterns.html](#) — SoA and Footnote Patterns.
- Study type, definition, terminology and the issue catalogue: [docs/phase0_definition_and_terms.md](#) .
- Protocol set (chosen / rejected): [docs/project_protocols.md](#) .
- Evidence: [sources/EMP_Study.docx](#) (notes + SoA screenshots, incl. NCT05469126 / J2A-MC-GZGM).
- Prior USDM tracing and SoA-shape findings: [docs/lessons_learned.md](#) .
- ICH M3(R2); FDA Exploratory IND Guidance (2006); EMA M3(R2).